KEMENTERIAN PERINDUSTRIAN
REPUBLIK INDONESIA

SMART ENGINEERING FOR A SUSTAINABLE FUTURE

Artificial Intelligence and Digital Technologies

Sekretaris Jenderal
Kementerian Perindustrian

Disampaikan pada:
World Engineering Day 2026
4 Maret 2026

Kementerian Perindustrian RI

TAHAP PEMBANGUNAN INDUSTRI JANGKA PANJANG RPJPN TAHUN 2025 - 2045

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TRANSFORMASI INDUSTRI
Berbasis SDA → Berbasis Inovasi dan Teknologi

Ketuntasan RPJPN 2005-2025	Penguatan Ekosistem Industrialisasi dan Peningkatan Kompleksitas Produk Industri	Peningkatan kompleksitas produk industri	Penguatan daya saing industri menuju ekspansi global	Indonesia *Manufacturing The World*
2024 Peranan Industri: **19,9-20,5%**	**2025-2029** Peranan Industri: **21,9%**	**2030-2034** Peranan Industri: **26,6%**	**2035-2039** Peranan Industri: **30,0%**	**2040-2045** Peranan Industri: **28,0%**
Fokus pada **perbaikan produktivitas dan daya saing**	**Penguatan struktur dan hilirisasi industri dan ekosistem industrialisasi** melalui investasi di sektor prioritas	**Peningkatan diversifikasi produk industri** yang mengarah *medium-high tech* berbasis sumber daya unggulan	**Peningkatan integrasi industri dalam negeri ke pasar global** secara masif dan agresif	**Indonesia sebagai pusat GVC global** dan menjadi pusat jasa manufaktur maju di tingkat regional

Sumber : UU No. 59 tahun 2024 tentang RPJPN tahun 2025-2045

TRANSFORMASI EKONOMI GLOBAL BERBASIS KECERDASAN BUATAN (AI)

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An Unprecedented Global Shift

The global economy is undergoing an **unprecedented transformation**. Artificial intelligence is no longer a distant vision, it is **rapidly reshaping how people live, work, and interact**, while redefining industries, economies, and societies worldwide.

AI as a New Engine of the Digital Economy

This AI-driven technological revolution is creating **new momentum for the digital economy**, driving productivity gains, redefining competitiveness, and unlocking **entirely new sources of economic and social value,** with an impact that may surpass previous technological revolutions.

From Option to Strategic Imperative

With **more than 70 countries** already adopting national AI strategies, **intelligent transformation is no longer optional,** it has become a **must-do agenda** to remain competitive in the global economy.

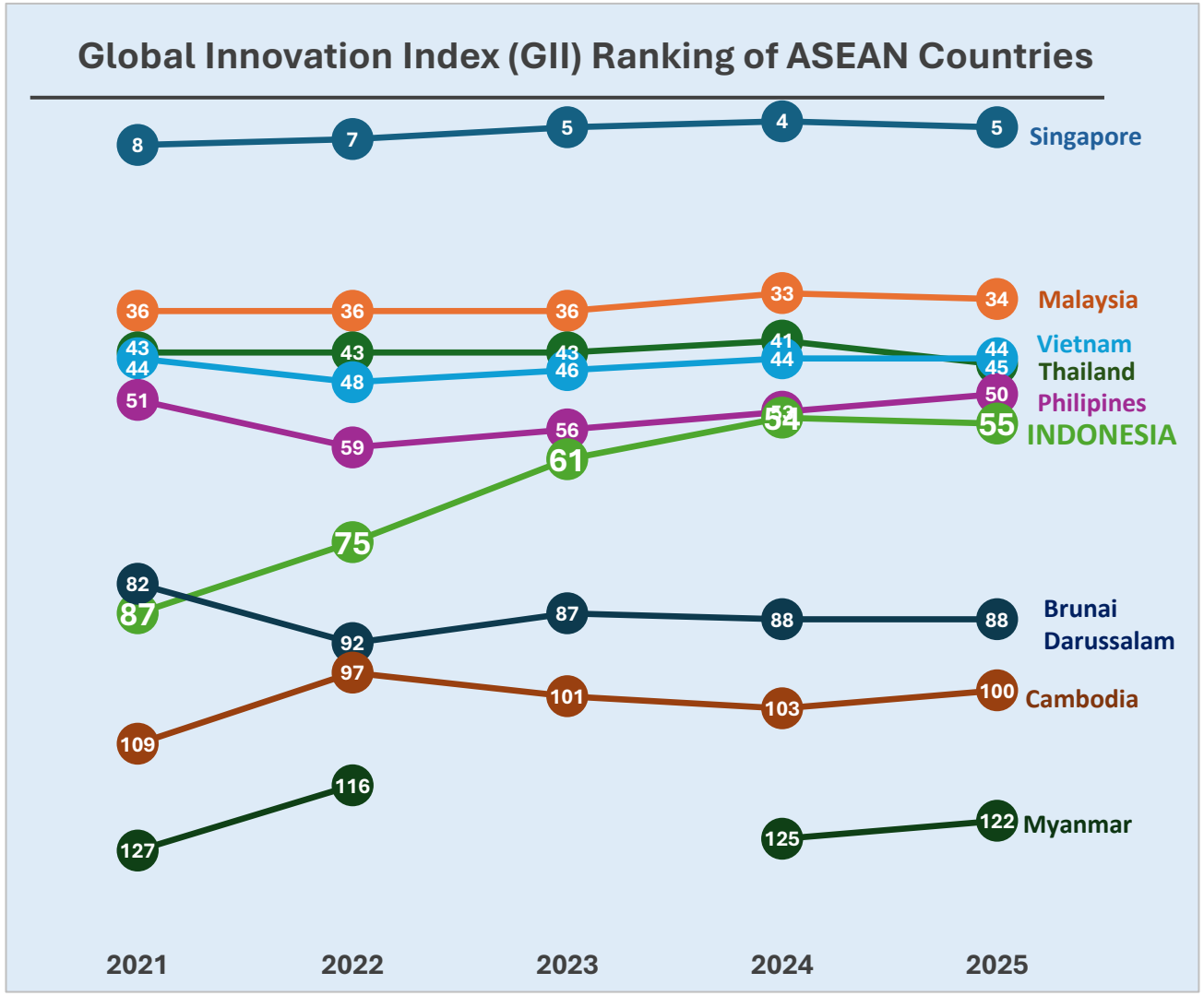
Besaran Dampak Ekonomi Dunia

US$22.3 trillion

3.7% of global GDP 2030

(Source: IDC)

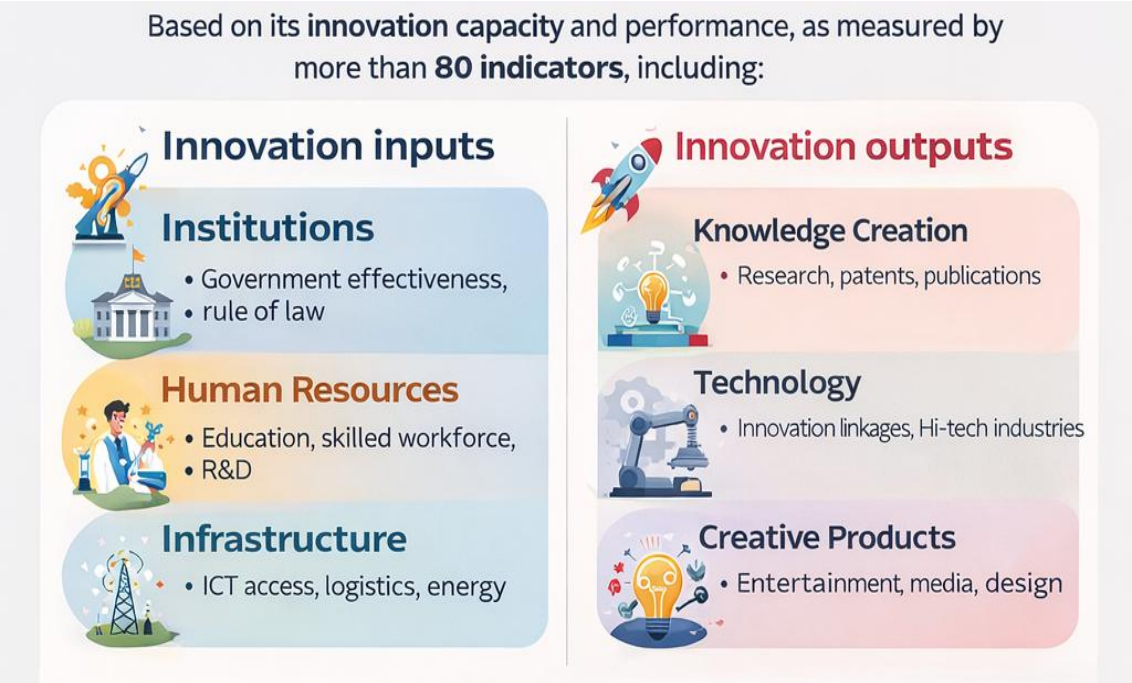
GLOBAL INNOVATION INDEX (GII) - ASEAN COUNTRY



Source : www.wipo.int (processed)

Global Innovation Index (GII)

Pada 2025 Indonesia menduduki peringkat 55 dari **139 negara**



GLOBAL DIGITALIZATION AND INTELLIGENCE INDEX (GDII)

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GDII Economies Coverage

High-Income (42 Economies)	
Economy Name	
Australia	Luxembourg
Austria	Netherlands
Bahrain	New Zealand
Belgium	Norway
Bulgaria	Oman
Canada	Poland
Chile	Portugal
Croatia	Qatar
Czech Republic	Romania
Denmark	Saudi Arabia
Estonia	Singapore
Finland	Slovak Republic
France	Slovenia
Germany	South Korea
Greece	Spain
Hungary	Sweden
Ireland	Switzerland
Italy	United Arab Emirates
Japan	United Kingdom
Kuwait	United States
Lithuania	Uruguay

Upper-Middle Income (24 Economies)	
Economy Name	
Algeria	Indonesia
Argentina	Iraq
Armenia	Kazakhstan
Azerbaijan	Malaysia
Botswana	Mexico
Brazil	Mongolia
China	Namibia
Colombia	Peru
Costa Rica	Serbia
Dominican Republic	South Africa
Ecuador	Thailand
Georgia	Türkiye

Lower-Middle Income & Low Income (24 Economies)	
Economy Name	
Bangladesh	Lebanon
Bolivia	Morocco
Cambodia	Nicaragua
Cameroon	Nigeria
DR Congo	Pakistan
Egypt	Philippines
Ghana	Tajikistan
Honduras	Tanzania
India	Tunisia
Jordan	Uganda
Kenya	Uzbekistan
Kyrgyz Republic	Vietnam

Source : Global Digitalization and Intelligence Index 2025 (www.Huawei.com)

GDII telah diterapkan untuk mengukur kesiapan digital **lebih dari 90 negara,** dan menilai bagaimana negara-negara tersebut bersiap untuk bersaing dan berkembang di era AI

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Making Indonesia 4.0

MAKING INDONESIA 4.0

Merupakan salah satu langkah penting pemerintah dalam mengakselerasi transformasi digital di Indonesia sebagai bentuk respon pemerintah terhadap fenomena Revolusi Industri 4.0. **Insinyur menjadi tonggak penting dalam penerapan making Indonesia 4.0. Insinyur menjadi eksekutor** dalam mendorong industri untuk **bertransformasi dan mengikuti perkembangan era digital**.

Aspirations:
Top 10 Global Economies in 2030

PREREQUISITES

- **10% net export contribution to GDP** (restore net export position to the same level as in 2000)
- **2x increase in productivity relative to cost** (increase production by managing costs similar to India)
- **2% R&D expenditure relative to GDP** (build local innovation capabilities at the same level as China)

ESTIMATED BENEFITS OF INDUSTRY 4.0 IMPLEMENTATION

- **GDP growth of 1-2% per annum from the baseline in 2030** (increase in real GDP growth from ± 5% to 6-7% YoY between 2018-2030)
- **Creation of over 10 million jobs by 2030** (increase in employment from ± 20 million to over 30 million by 2030)
- **Manufacturing contribution of over 25% to GDP by 2030** (Increase in manufacturing contribution to GDP from ± 16% to 25% by 2030)

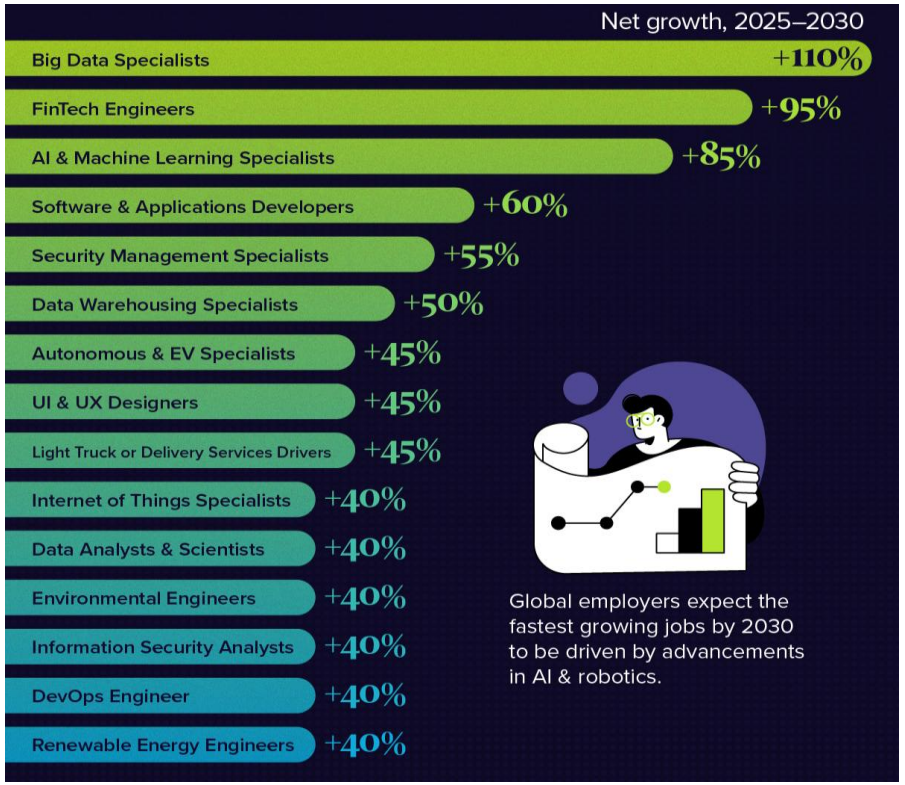
PRIORITY SECTORS

Food & Beverages | Automotive | Chemicals | Textiles and Textile Products | Electronics | Pharmaceuticals | Medical Devices

PRIORITY STRATEGIES

1. Improvement of Material Flow Systems
2. Redesigning industrial zones
3. Accommodating sustainable standards
4. Empowerment of MSMEs
5. Building digital infrastructure
6. Attracting foreign investment
7. Improving human resource quality
8. Formation of an innovation ecosystem
9. Implementation of technology investment incentives
10. Harmonization of regulations and policies

Cross-Ministry/Agency Collaboration



Source : World Economic Forum – Future Jobs Report 2025 (diolah)

Tantangan nyata bagi seluruh insinyur Indonesia adalah untuk memenuhi kebutuhan **kualifikasi terhadap kemampuan berbasis digital**. Berdasarkan data World Economic Forum dalam Future Jobs Report 2025, pada masa mendatang, dunia akan membutuhkan banyak insinyur yang bergerak pada bidang digitalisasi.

Source: Making Indonesia 4.0 Roadmap Document, 2018

DRIVING I4.0 : PROGRAM STRATEGIS KEMENTERIAN PERINDUSTRIAN

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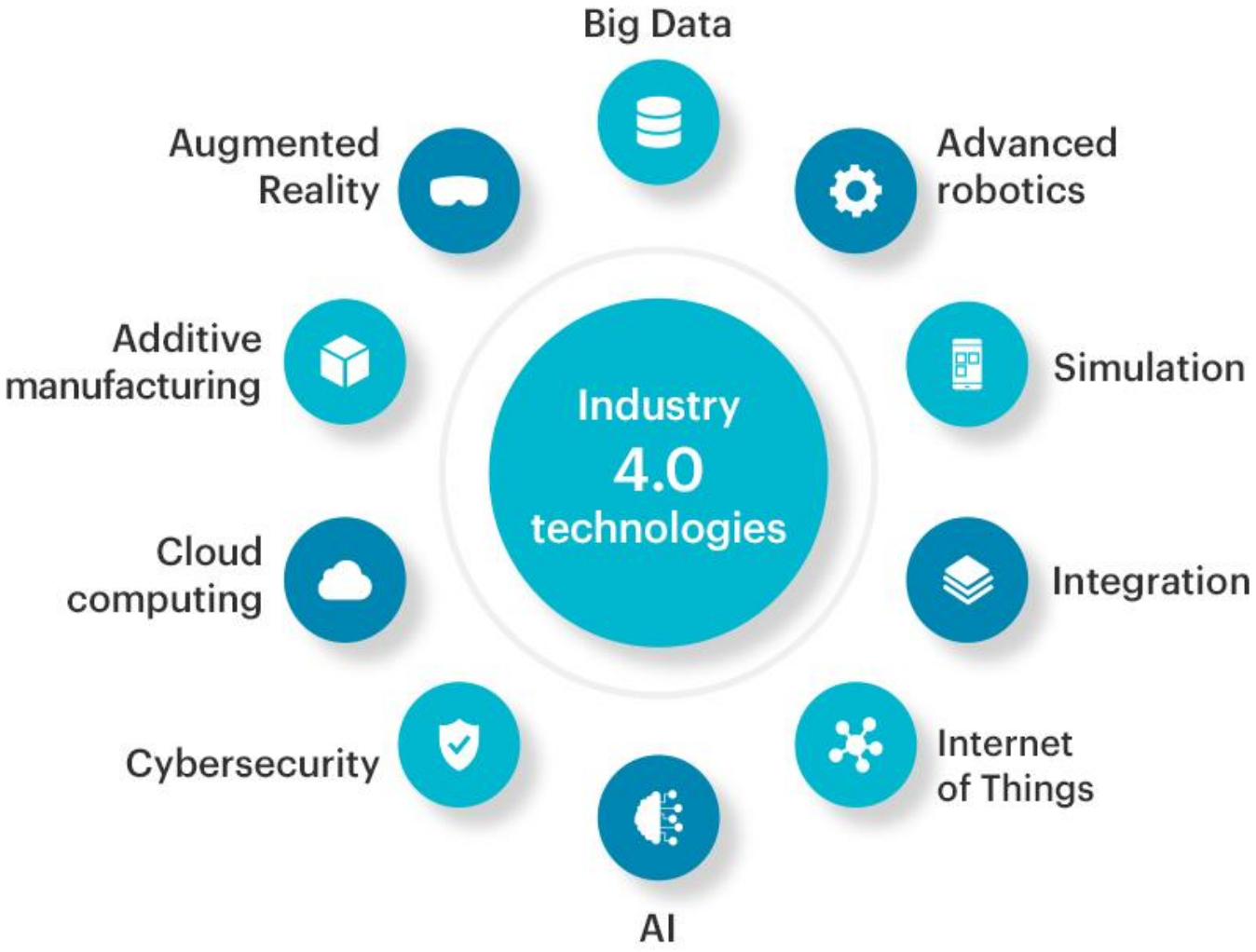


TRANSFORMASI DIGITAL PENINGKATAN DAYA SAING INDUSTRI

Kementerian Perindustrian REPUBLIK INDONESIA

29 Perusahaan National Lighthouse Industri 4.0 :

Industri Kimia

- PT Pupuk Kalimantan Timur
- PT Pupuk Petrokimia Gresik
- PT Pupuk Sriwijaya Palembang
- PT Pupuk Indonesia (Persero)
- PT Candra Asri Pacific

Industri Farmasi

- PT Kalbe Farma
- PT Paragon Technology and Innovation

Industri Tekstil

- PT Globalindo Intimates
- PT Prima Sejati Sejahtera
- PT Asia Pacific Rayon
- PT Pancaprima Ekabrothers

Industri Semen

- PT Semen Tonasa
- PT Solusi Bangun Indonesia

Industri Aneka

- PT Mattel Indonesia

Industri Elektronika

- PT Schneider Electronic Manufacturing Batam
- PT Yamaha Electronics Manufacturing Indonesia
- PT Hartono Istana Teknologi
- PT Infineon Technologies Batam

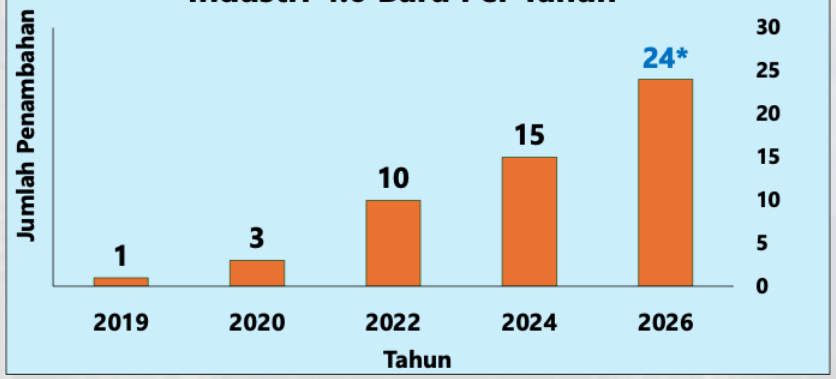
Industri Otomotif

- PT Akebono Brake Astra Indonesia
- PT Astra Honda Motor
- PT Astra Komponen Indonesia
- PT Aisin Indonesia Automotive
- PT Toyota Motor Manufacturing Indonesia

Industri Makanan dan Minuman

- PT Indolakto
- PT Gelora Djaja
- PT Kalbe Morinaga Indonesia dan PT Shanghiang Perkasa
- PT Amerta Indah Otsuka
- PT Tirta Investama
- PT Lautan Natural Krimerindo

Penambahan National Lighthouse Industri 4.0 Baru Per Tahun



*PMA

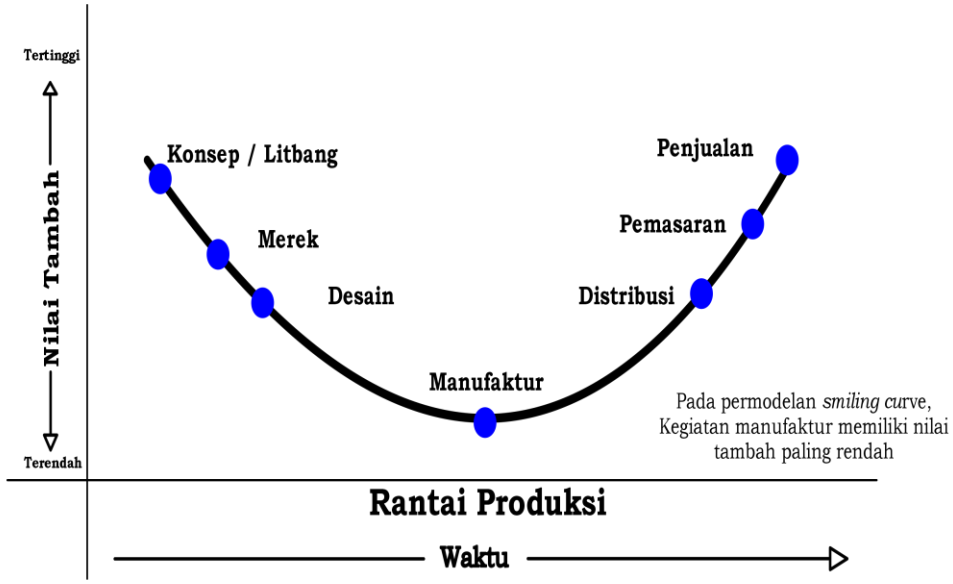
Dampak transformasi digital terhadap 29 Perusahaan National Lighthouse Industri 4.0:

- **Speed-to-Market** up to 600%
- **Agility** up to 50%
- **Productivity** up to 101%
- **Financial** up to 200%
- **Customer Experience** up to 97%
- **Sustainability** up to 190%

LINGKUP PERAN INSINYUR DALAM RANTAI PRODUKSI

Sektor Perindustrian

Insinyur memegang peran krusial guna mendorong kontribusi sektor Perindustrian terhadap PDB Nasional. Insinyur bertindak sebagai penggagas dari **inovasi, peningkat efisiensi, dan pengembangan industri berkelanjutan**. Insinyur adalah pemecah masalah yang strategis dan inovator yang tak kenal lelah, yang memastikan sektor perindustrian terus berkembang, lebih efisien, aman, dan kompetitif di era modern.



Insinyur berada pada seluruh aktivitas rantai industri dari awal hingga akhir produksi

Contoh Peran Insinyur dalam Aktivitas Industri

Engineering, Procurement and Construction

Insinyur berperan penting dalam proses Engineering Design, Penentuan Mesin Peralatan, hingga pembangunan Gedung industri

Desain Industri

Insinyur berperan penting dalam proses Engineering Design produk industri otomotif, tekstil, dan produk industri lainnya

Process Engineering

Insinyur berperan penting dalam pelaksanaan aktivitas produksi di Industri pengolahan

Inspection & Quality

Insinyur berperan penting dalam pelaksanaan pengawasan terhadap seluruh kegiatan industri

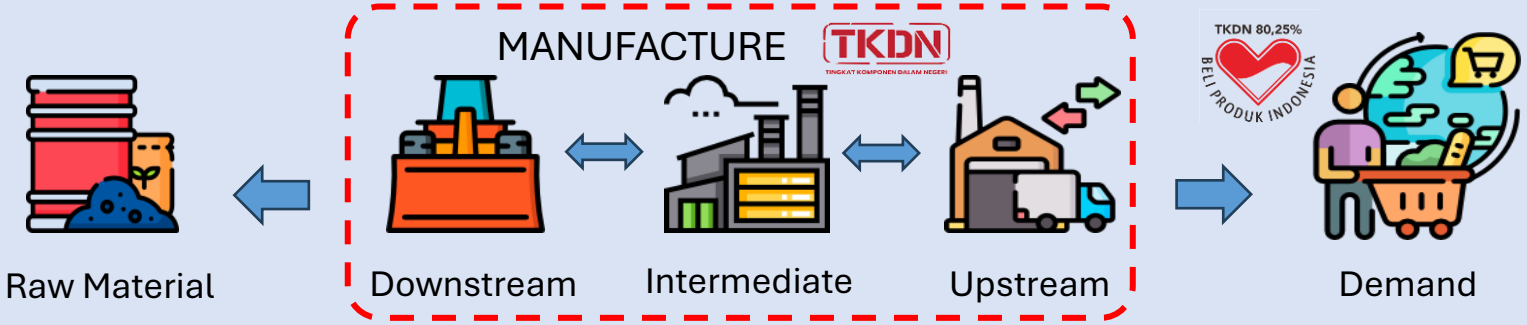
PERAN INSINYUR DALAM MENUNJANG KEBIJAKAN PEMERINTAH

Kementerian Perindustrian
REPUBLIK INDONESIA

TKDN (Tingkat Komponen Dalam Negeri)

Pemerintah Republik Indonesia melalui Kementerian Perindustrian telah menetapkan **kebijakan untuk mewajibkan seluruh pembelanjaan barang dan jasa di lingkungan pemerintah guna mendorong peningkatan daya saing industri nasional.** Insinyur menjadi salah satu penunjang utama dalam meningkatkan nilai TKDN Barang, Jasa, maupun Gabungan Barang dan Jasa.

JENIS-JENIS TKDN



Komponen TKDN Barang

Saat ini Kemenperin sedang dilakukan penyusunan peraturan **TKDN Jasa Industri** yang nantinya akan **menitikberatkan pada Sumber Daya Manusia Industri** sebagai salah satu komponen dalam perhitungan NilaiTKDN

Multiplier Effect on the Economy

Every Rp1 spent on domestic products generates Rp2.2 in economic activity.

Industrial Growth

Industrial sectors whose products are purchased by the government and state-owned enterprises experienced significantly higher growth.

Employment Expansion

There has been an increase in employment absorption (Data of 2023–2024)

STRATEGI PENGUATAN PERAN INSINYUR

Kementerian Perindustrian
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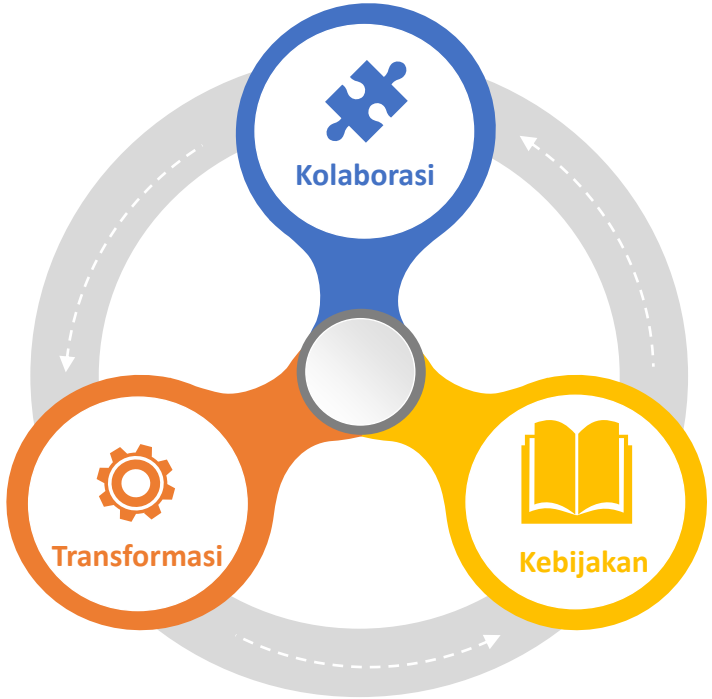
Kolaborasi, Kebijakan, Transformasi



Transformasi Digital

Penerapan teknologi industri 4.0 untuk meningkatkan daya saing dan efisiensi nasional.

Kolaborasi Insinyur - Industri-Universitas

Penguatan riset terapan, magang industri, kurikulum berbasis kebutuhan industri.

Kebijakan dan Regulasi

Pemerintah perlu menetapkan standar sertifikasi nasional dan mendukung insentif industri serta insinyur sehingga meningkatkan daya saing sektor Perindustrian.

“A strong industry is one with innovative engineers capable of solving various global issues to support the development of the national industry.”
Collaboration among engineers in the industrial sector will significantly help achieve the Industrial Law Draft of making Indonesia a global industrial power. The government, together with engineers and industry, can build a solid foundation for developing a sustainable, competitive technology-based industry.

TERIMA KASIH

KEMENTERIAN PERINDUSTRIAN

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